

Test Report

Electromagnetic Compatibility (EMC)

Report No. : HA0126NB03034601EM
Applicant : Ningbo Golong Robot Technology Co., Ltd
Address : No.2 yunqiao east road, Binhai New Area, Southern Ningbo, Ninghai, Ningbo, Zhejiang, China
Trade Mark(s) : --
Manufacturer : Same as the applicant
Address : Same as the applicant
Factory(ies) : Same as the applicant
Address : Same as the applicant

Equipment Under Test (EUT):

EUT Name : Robot Vacuum Cleaner
Model/Type No. : A801, A801x, A802, A802x(x=a-z)
Standards : Refer to page 2
Date of Receipt : March 23, 2026
Date of Test : March 24, 2026 to April 15, 2026
Date of Issue : April 16, 2026
Test Result : **PASS***

Prepared By:

Ningbo HATEK Co., Ltd.

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Prepared By

Leo Lin

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Engineer

Reviewed By



Bin Dong

Technical Manager

*The test results have been reviewed against the Directives above and found to meet their essential requirement. The results shown in this test report refer only to the sample(s) tested. This document cannot be reproduced except in full, without prior written approval of HATEK.

1 Test Summary

1.1 Test Items

No.	Test Items	Result
1.	Disturbance Voltage on Mains Terminal 0.15MHz- 30MHz	P
2.	Discontinuous Disturbance Voltage/Click	N/A
3.	Continuous Disturbance Power, 30MHz - 300MHz	N/A
4.	Radiation Emission, 30MHz - 1000MHz	P
5.	Harmonic Current	P
6.	Voltage Fluctuations-Flicker	P
7.	ESD	P
8.	Radiated Immunity (80MHz - 1GHz)	P
9.	Electrical Fast Transients (EFT)	P
10.	Surge Immunity	P
11.	Injected Currents, 0.15MHz - 80MHz	P
12.	Voltage Dips and Interruptions	P
Remark:		P: Pass/ F: Fail/ N/A: Not Applicable

1.2 Test Specification

The equipment(s) comply with the requirements according to the following standards:

EN IEC 55014-1:2021: Electromagnetic compatibility-Requirements for household appliances, electric tools and similar apparatus Part 1: Emission;

EN IEC 55014-2:2021: Electromagnetic compatibility-Requirements for household appliances, electric tools and similar apparatus Part 2: Immunity;

EN IEC 61000-3-2:2019+A1:2021+A2:2024: Electromagnetic compatibility (EMC) - Part 3-2: Limits - Limits for harmonic current emissions (equipment input current ≤16 A per phase);

EN 61000-3-3:2013+A1:2019+A2:2021 : Electromagnetic compatibility (EMC) - Part 3-3: Limits - Limitation of voltage changes, voltage fluctuations and flicker in public low-voltage supply systems, for equipment with rated current ≤ 16 A per phase and not subject to conditional connection.

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2 General Information

2.1 Client Information

Applicant : Ningbo Golong Robot Technology Co., Ltd
Address : No.2 yunqiao east road, Binhai New Area, Southern Ningbo, Ninghai,
Ningbo, Zhejiang, China

2.2 General Description of E.U.T.

Electrical Rating : DC 19V by Adapter Input:100-240V, 50/60Hz; Output:19V, 0.6A
DC 11.1V by Battery
Highest internal frequency (Fx): < 15MHz

2.3 Identifies and differences:

All models are the same, just the model names are different.
Therefore, we test A801 and the worst test data is listed in the report as representative.

2.4 Description of Support Units

Description	Manufacturer	Model No.	Ratings
--	--	--	--
--	--	--	--

2.5 Mode of operation during the test

Within this test report, EUT was tested under all available operation modes and tested under its rating voltage and frequency. Other voltage and frequency are specified if used.

2.6 Environment

- ☒ Residential (domestic) environment
- ☒ Commercial and light-industrial environment
- ☐ Industrial environment
- ☐ Medical environment.

2.7 Submitted Documents

Constructional Data Form for EMC
Circuit diagram, user's manual, labels and construction drawings etc.

3 Test Facility and Instrument list

3.1 Test Facility

Laboratory : Ningbo HATEK Co., Ltd.

Address : 1F, Building 9-1, Sino-German Industrial Park, No.502, Xinxing San Road, Hi-tech Zone, Cixi City, Ningbo, Zhejiang, 315300, China

The used test equipment is in accordance with CISPR 16-1 series standards for measurement of radio interference.

3.2 Instrument list

Table 1: List of Test and Measurement Equipment of Laboratory

No.	Equipment	Manufacturer	Model	Serial No.	Due Date
Mains Terminal Continuous Disturbance Voltage					
1.	EMI Test Receiver	R&S	ESR3	100050	2026/12/29
2.	Artificial Mains Network	Schwarzbeck	ENV216	101526	2026/12/29
3.	ISN	Schwarzbeck	NTFM8158	08542509	2026/12/29
4.	Shielded Room	Taihe MaoRui	9m*5m*3.5m	--	2030/09/26
Harmonics & Flicker					
1.	Harmonic and Flicker Analyzer	ZESZIMMER	LMG641	08542509	2026/12/29
Click					
1.	Click Analyzer	AFJ	DDA55	14041843145	2026/12/29
2.	Artificial Mains Network	Schwarzbeck	ENV216	101526	2026/12/29
Disturbance Power					
1.	Absorbing Clamp	Schwarzbeck	MDS 21	--	2026/12/29
Radiated Emission					
1.	EMI Test Receiver	R&S	ESR7	--	2026/12/29
2.	Broadband Preamplifier	yian	TRLA—001060G30B	25072901	2026/12/29
3.	Broadband Antenna	Schwarzbeck	VULB 9168	01191	2026/12/29
4.	Horn Antenna	Schwarzbeck	BBHA 9120 D	--	2026/12/29
5.	Anechoic Chamber	Taihe MaoRui	9m x 6mx 6m	--	2030/09/26
Electrostatic Discharge (ESD) immunity					
1.	ESD Simulator	LIONCEL	ESD-203G	ESD203G-0250704	2026/05/20
Radio frequency electromagnetic field immunity (R/S) test					
1.	Signal generator	R&S	SMB 100A	--	2026/12/29
2.	Composite antenna	SRD	STLP 9129 pluS	--	2026/12/29
3.	Power amplifier	RFLIGHT	NTWPA-00810200	--	2026/12/29
Electrical Fast Transients (EFT) immunity					
1.	IMU 4000 immunity test system	EMC PARTNER GA	IMU4000 F5-S-D-V	106754-2020	2026/12/29
Surge immunity					
1.	IMU 4000 immunity test system	EMC PARTNER GA	IMU4000 F5-S-D-V	106754-2020	2026/12/29
Injected Currents immunity					
1.	Conducted Immunity Test System	ETEST	ES-608	251015	2026/12/29

No.	Equipment	Manufacturer	Model	Serial No.	Due Date
2.	6dB Attenuator	ETEST	JTH-SJ-100W-6dB	JTH20250312 002	2026/12/29
3.	CDN-M2	ETEST	ES-CDN-M2A	251016	2026/12/29
4.	CDN-M3	ETEST	ES-CDN-M3A-LNPE	251017	2026/12/29
5.	Electromagnetic Injection Clamp	ETEST	ES-TS120	251023	2026/12/29
Voltage Dips and Interruptions					
1.	IMU 4000 immunity test system	EMC PARTNER GA	IMU4000 F5-S-D-V	106754-2020	2026/12/29
2.	Harmonic and Flicker Analyzer	ZESZIMMER	LMG641	08542509	2026/12/29

3.3 Measurement Uncertainty

Conduction Emission (9 kHz-150 kHz)	:	U = 3.0 dB
Conduction Emission (150 kHz-30 MHz)	:	U = 2.6 dB
Disturbance Power (30 MHz-300 MHz)	:	U = 3.3 dB
Radiated Emission (30 MHz-1 GHz)	:	U = 4.2 dB
Radiated Emission (1 GHz-6 GHz)	:	U = 4.12 dB
Harmonic test	:	2.46%

Note: This uncertainty represents an expanded uncertainty expressed at approximately the 95% confidence level using a coverage factor of k=2.

4 Test Results EMISSION

4.1 Emission in the Frequency Range from 0 kHz to 30 MHz

4.1.1 Harmonics on AC Mains

General test information

Temperature	: 25°C
Relative Humidity	: 51 %RH
Test procedure	: EN IEC 61000-3-2:2019+A1:2021+A2:2024
Test duration	: 2.5min
Harmonic order	: 2 – 40 th
Frequency range	: 0 – 2kHz
Test result	: Pass

Block Diagram of Test Set up



Test Procedure

The harmonics on AC Mains in the frequency from 0 to 2 kHz were measured in accordance with EN IEC 61000-3-2:2019+A1:2021+A2:2024.

The measurement of Harmonics of the fundamental current was measured up to 40 order harmonics using a digital power meter with an analogue output and frequency analyzer which was integrated in the harmonic & flicker test system. This equipment is in compliance with the requirements of EN IEC 61000-3-2:2019+A1:2021+A2:2024.

The results indicated in the following tables and figures were those measured and recorded by an automatic measuring system.

Test Results:

According to the Clause 7 in the EN IEC 61000-3-2:2019+A1:2021+A2:2024

For the following categories of equipment, limits are not specified in this standard:

– equipment with a rated power of 75 W or less, other than lighting equipment;

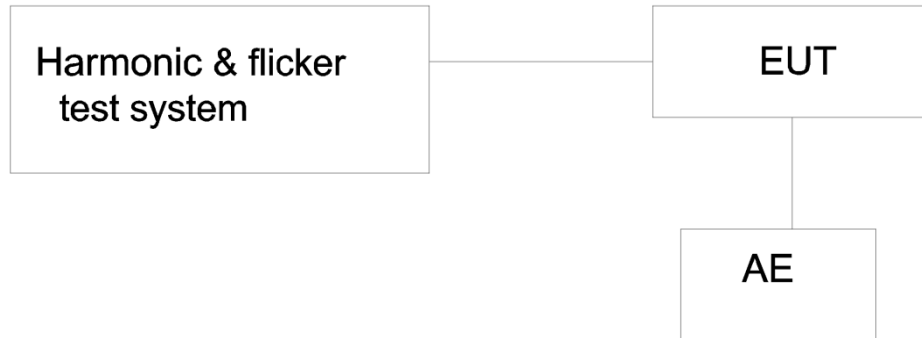
This product is not defined as lighting equipment, and has actual measured power less than 75W, therefore, no limit applies according to EN IEC 61000-3-2:2019+A1:2021+A2:2024.

4.1.2 Voltage changes, voltage fluctuations and flicker on AC mains

General test information

Test procedure	:	EN 61000-3-3:2013+A1:2019+A2:2021
Temperature	:	25°C
Relative Humidity	:	51 %RH
Test result	:	Pass

Block Diagram of Test Set up



Test Procedure

According to the A.10 of the EN 61000-3-3:2013+A1:2019+A2:2021:

According to the characteristics of the sample, as specified by clause 5 of the basic standard, following limits apply:

- the value of P_{st} shall not be greater than 1.0;
- the value of P_{lt} shall not be greater than 0.65;
- the value of $d(t)$ during a voltage change shall not exceed 3.3% for more than 500ms;
- the relative steady-state voltage change, d_c , shall not exceed 3.3%;
- the maximum relative voltage change d_{max} , shall not exceed 4%.

The measurement was carried in accordance with Annex B of the basic standard and the EUT was set to produce the most unfavourable sequence of voltage changes. Following are the measurement results obtained via an automatic testing system.

Table 2: Voltage fluctuations and flicker measurement results

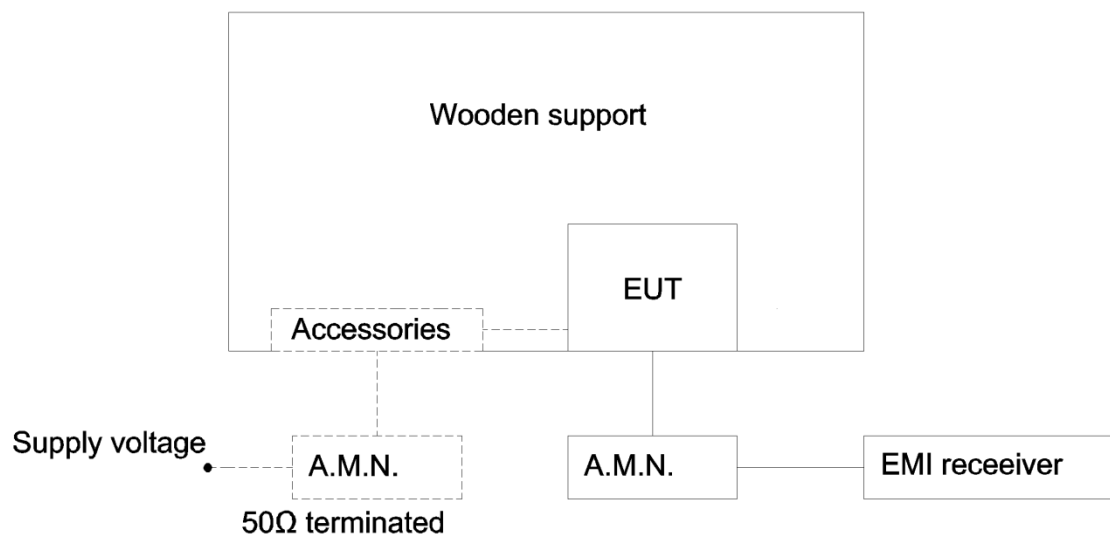
	d_c	$d_{max}(average)$	$d(t)$	P_{st}	P_{lt}
Limits	3.3%	4%	3.3%/500ms	1.0	N/A
Result	0.05%	0.05%	0.00ms	0.011	-

4.1.3 Mains Terminal Continuous Disturbance Voltage

General test information

Test procedure	: EN IEC 55014-1:2021 and CISPR 16-1 series standards
Frequency range	: 0.15-30MHz
Kind of test site	: EMC Chamber
Temperature	: 25 °C
Relative Humidity	: 51 %RH
Operational condition	: Charging mode
Artificial hand	: No
Earthing	: No
Test result	: Pass

Block Diagram of Test Set up



- ☐ For table top equipment, wooden support is 0.8m height.
- ☒ For floor standing equipment, wooden support is 0.1m height.

Test Procedure

The measurement setup was made according to EN IEC 55014-1:2021 in an EMC Chamber.

Prior to the measurements the test object operated about 15 minutes (warm-up) in order to stabilize its operating conditions and to ensure reliable measurement values.

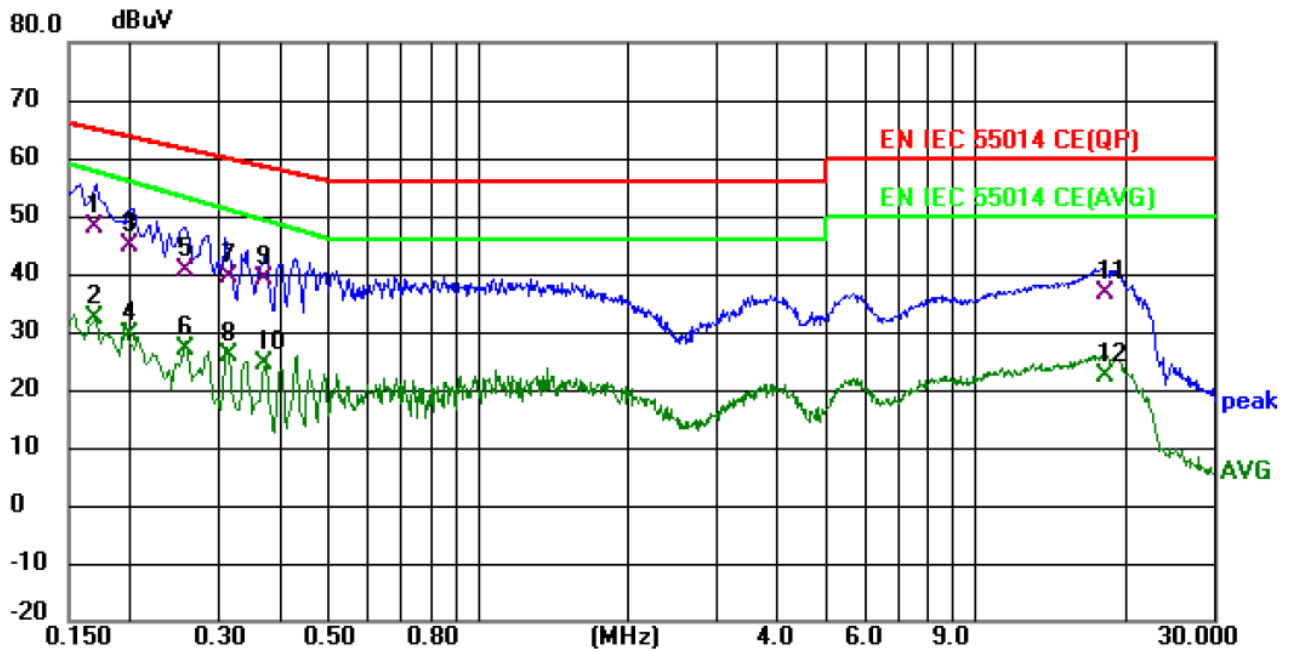
Furthermore, an internal calibration with the test receiver was conducted prior to each measurement. And the measurement was made in the state the maximum disturbance was obtained.

The tested object was set-up on a wooden table. The length of the power cord of the tested object was about 1.5m. The EUT was set 0.8m away from the AMN. The cord longer than necessary to be connected to the AMN was folded forth and back parallel so as to form a bundle with a length between 0.3m and 0.4m. The EUT (Equipment under Test) was wrapped with artificial hand that was earthed through the Artificial Mains Network (AMN).

The Interference Voltage was determined according to clause 5 of EN IEC 55014-1:2021 while measuring the line and neutral conductor by turns.

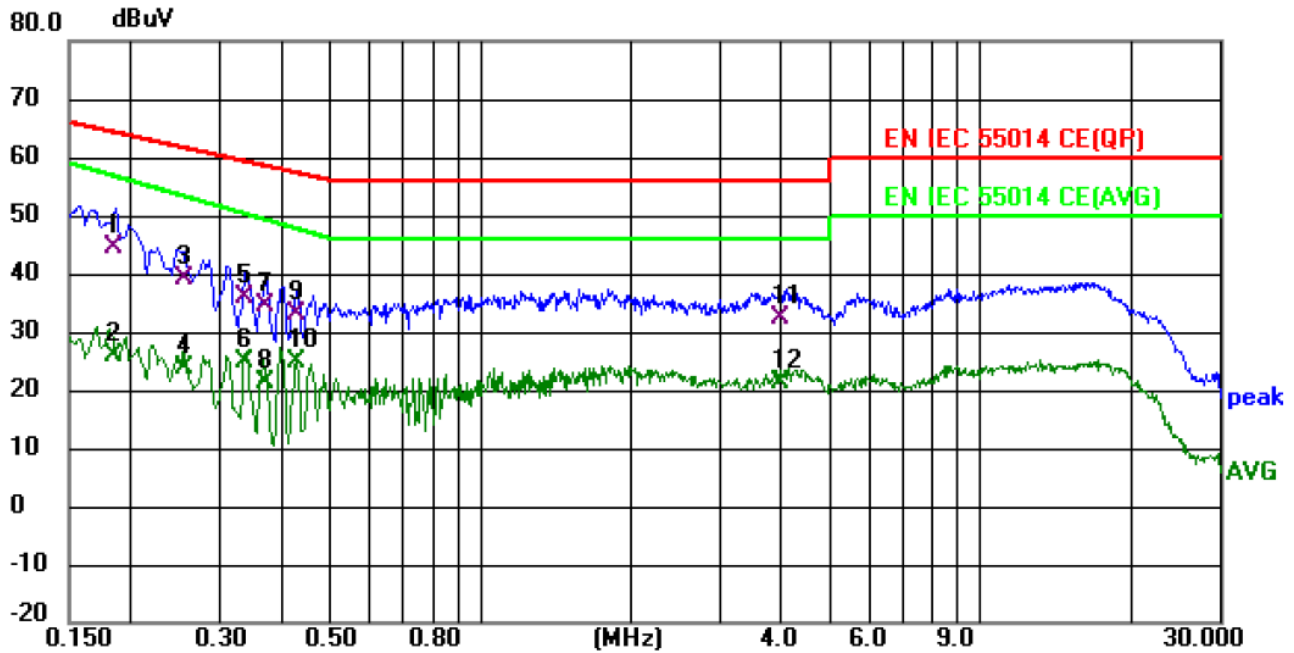
In the Figures, the symbol “+” means Quasi-Peak Value and the symbol “x” means Average Value which was measured in final measurement.

Figure 1: Test Curve of Conducted Emission in the frequency range of 150kHz - 30MHz, L line (120V)



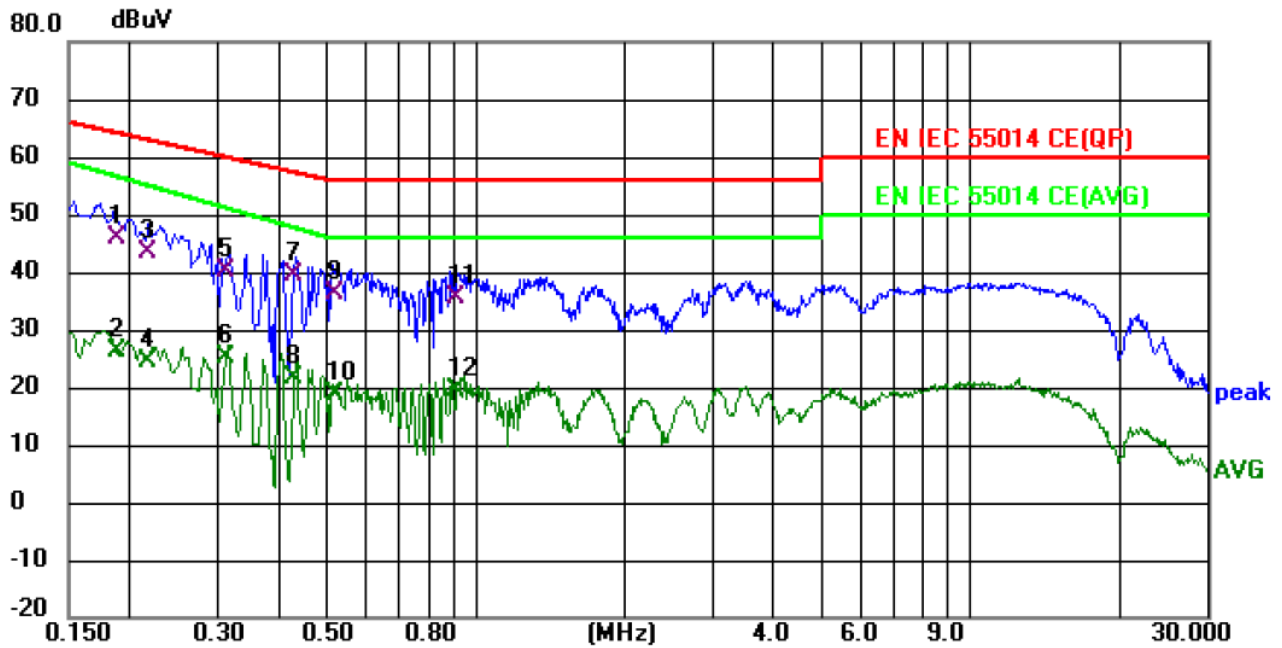
No.	Frequency (MHz)	Reading (dBuV)	Factor (dB)	Level (dBuV)	Limit (dBuV)	Margin (dB)	Detector	P/F	Remark
1 *	0.169	38.59	9.53	48.12	65.01	-16.89	QP	P	
2	0.169	22.73	9.53	32.26	57.72	-25.46	AVG	P	
3	0.199	35.43	9.54	44.97	63.65	-18.68	QP	P	
4	0.199	19.99	9.54	29.53	55.95	-26.42	AVG	P	
5	0.258	31.06	9.55	40.61	61.50	-20.89	QP	P	
6	0.258	17.37	9.55	26.92	53.14	-26.22	AVG	P	
7	0.313	29.98	9.55	39.53	59.89	-20.36	QP	P	
8	0.313	16.18	9.55	25.73	51.06	-25.33	AVG	P	
9	0.372	29.42	9.56	38.98	58.46	-19.48	QP	P	
10	0.372	14.77	9.56	24.33	49.19	-24.86	AVG	P	
11	18.136	26.50	9.96	36.46	60.00	-23.54	QP	P	
12	18.136	12.45	9.96	22.41	50.00	-27.59	AVG	P	

Figure 2: Test Curve of Conducted Emission in the frequency range of 150kHz - 30MHz, N line (120V)



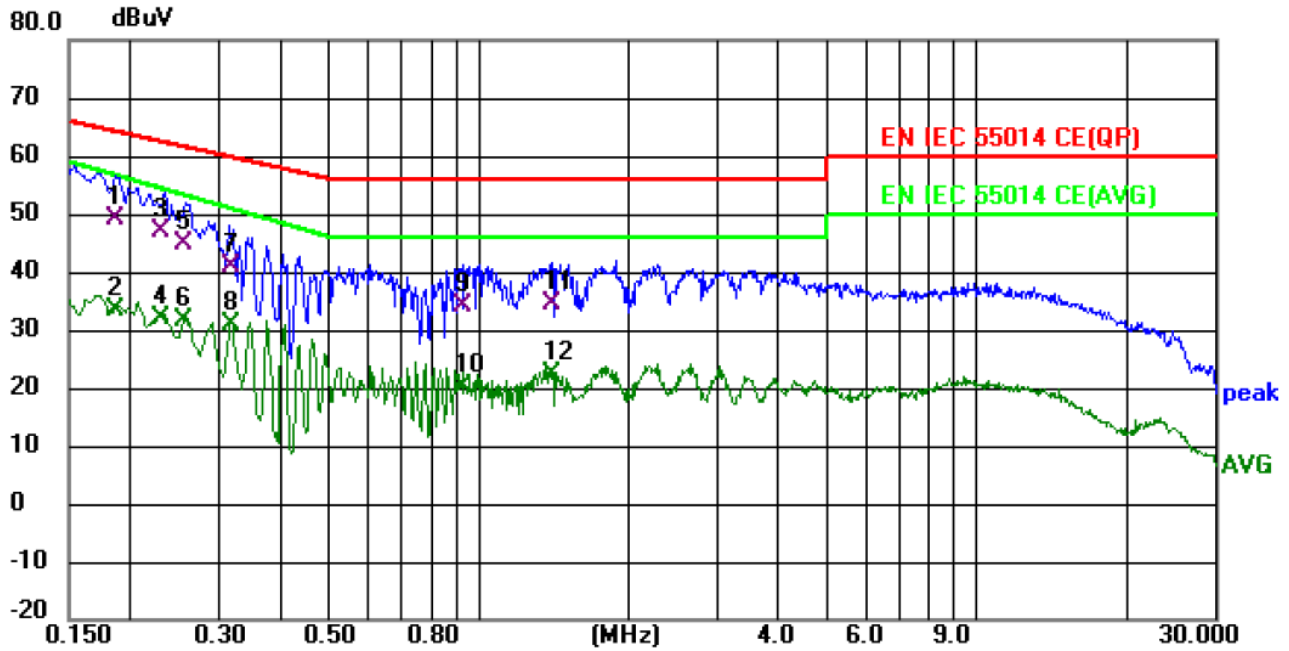
No.	Frequency (MHz)	Reading (dBuV)	Factor (dB)	Level (dBuV)	Limit (dBuV)	Margin (dB)	Detector	P/F	Remark
1 *	0.183	35.03	9.53	44.56	64.35	-19.79	QP	P	
2	0.183	16.49	9.53	26.02	56.85	-30.83	AVG	P	
3	0.254	29.56	9.55	39.11	61.63	-22.52	QP	P	
4	0.254	14.34	9.55	23.89	53.31	-29.42	AVG	P	
5	0.337	26.18	9.55	35.73	59.28	-23.55	QP	P	
6	0.337	15.30	9.55	24.85	50.26	-25.41	AVG	P	
7	0.372	24.79	9.56	34.35	58.46	-24.11	QP	P	
8	0.372	11.66	9.56	21.22	49.19	-27.97	AVG	P	
9	0.426	23.63	9.56	33.19	57.33	-24.14	QP	P	
10	0.426	15.21	9.56	24.77	47.73	-22.96	AVG	P	
11	4.002	22.68	9.68	32.36	56.00	-23.64	QP	P	
12	4.002	11.64	9.68	21.32	46.00	-24.68	AVG	P	

Figure 3: Test Curve of Conducted Emission in the frequency range of 150kHz - 30MHz, L line (230V)



No.	Frequency (MHz)	Reading (dBuV)	Factor (dB)	Level (dBuV)	Limit (dBuV)	Margin (dB)	Detector	P/F	Remark
1	0.187	36.37	9.53	45.90	64.17	-18.27	QP	P	
2	0.187	16.83	9.53	26.36	56.62	-30.26	AVG	P	
3	0.216	33.78	9.54	43.32	62.97	-19.65	QP	P	
4	0.216	14.98	9.54	24.52	55.06	-30.54	AVG	P	
5	0.311	30.53	9.55	40.08	59.94	-19.86	QP	P	
6	0.311	15.72	9.55	25.27	51.13	-25.86	AVG	P	
7 *	0.429	29.81	9.56	39.37	57.27	-17.90	QP	P	
8	0.429	12.05	9.56	21.61	47.65	-26.04	AVG	P	
9	0.515	26.81	9.56	36.37	56.00	-19.63	QP	P	
10	0.515	9.17	9.56	18.73	46.00	-27.27	AVG	P	
11	0.917	25.79	9.60	35.39	56.00	-20.61	QP	P	
12	0.917	9.73	9.60	19.33	46.00	-26.67	AVG	P	

Figure 4: Test Curve of Conducted Emission in the frequency range of 150kHz - 30MHz, N line (230V)



No.	Frequency (MHz)	Reading (dBuV)	Factor (dB)	Level (dBuV)	Limit (dBuV)	Margin (dB)	Detector	P/F	Remark
1 *	0.186	39.73	9.53	49.26	64.21	-14.95	QP	P	
2	0.186	23.76	9.53	33.29	56.68	-23.39	AVG	P	
3	0.231	37.57	9.55	47.12	62.41	-15.29	QP	P	
4	0.231	22.28	9.55	31.83	54.34	-22.51	AVG	P	
5	0.257	35.19	9.55	44.74	61.53	-16.79	QP	P	
6	0.257	22.23	9.55	31.78	53.19	-21.41	AVG	P	
7	0.319	31.32	9.55	40.87	59.73	-18.86	QP	P	
8	0.319	21.28	9.55	30.83	50.85	-20.02	AVG	P	
9	0.930	24.47	9.60	34.07	56.00	-21.93	QP	P	
10	0.930	10.53	9.60	20.13	46.00	-25.87	AVG	P	
11	1.389	24.76	9.61	34.37	56.00	-21.63	QP	P	
12	1.389	12.71	9.61	22.32	46.00	-23.68	AVG	P	

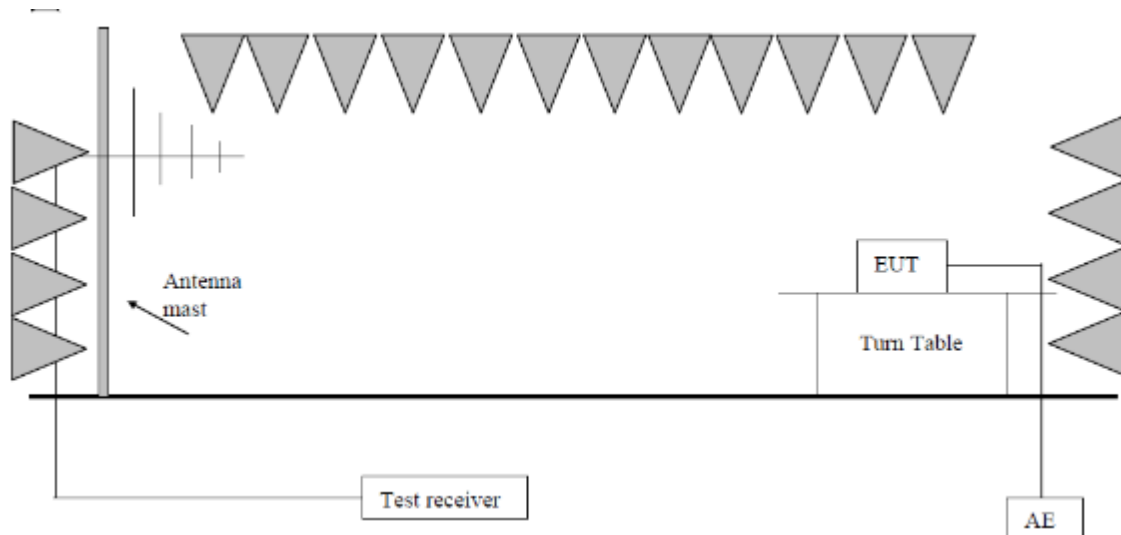
4.2 Emission in the Frequency Range from 30 MHz to 1000 MHz

4.2.1 Radiated disturbance

General test information

Frequency Range	: 30 – 1000MHz
Kind of test site	: EMC Chamber
Temperature	: 25 °C
Relative Humidity	: 51 %RH
Operational condition	: Normal operation and/or charging
Port	: Mains
Limit	: EN IEC 55014-1:2021, Table 3, Household and similar appliances
Test result	: Pass

Block Diagram of Test Set up



- ☒ For table top equipment, wooden support is 0.8m height.
- ☐ For floor standing equipment, wooden support is 0.1m height.

Measuring configuration and description

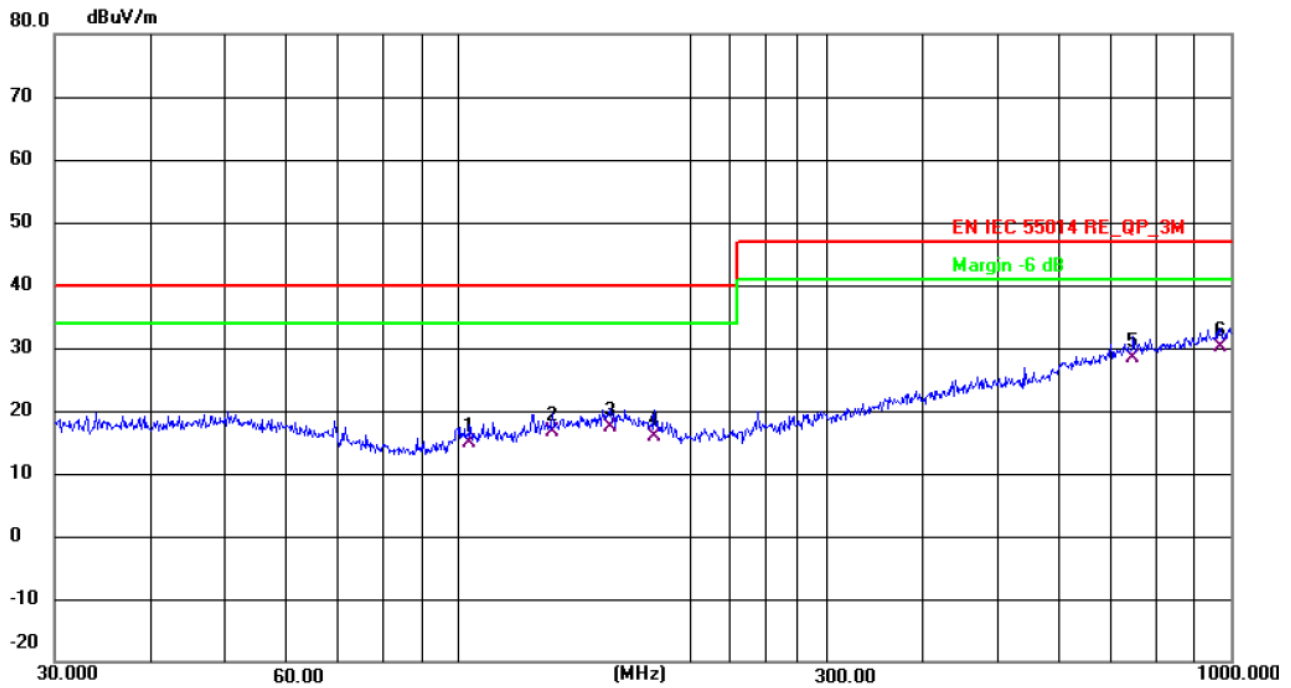
The measurement was applied in a semi-anechoic chamber. Measurement was performed according to CISPR 22. Setting of EUT is according to clause 7 of EN 55014-1. The bandwidth setting on Test Receiver was 120kHz. The frequency range from 30MHz to 1000MHz was checked.

The radiated disturbance test was performed in a 3m semi-anechoic chamber. The test distance is 3m. The 10m radiated emission limits are converted to 3m radiated emission limits by an inverse proportionality of 20 dB per decade. The normalized site attenuation of the semi-anechoic chamber is regularly calibrated to ensure the radiated disturbance test results are valid. During the test, the EUT was placed on a 0.8m high wooden support above the reference ground plane. The turntable was rotated 360° around and the antenna was varied from 1m to 4m to find the maximum disturbance. The test was performed with the antenna both in its horizontal and vertical polarizations.

The following figures were those measured and recorded by a test receiver. The curves in the figure were those measured with a Peak detector. The symbol "X" in the figures is those of QP value which were measured in final measurement. Quasi-peak measurements were only performed at those critical frequencies obtained during the test with Peak Detector.

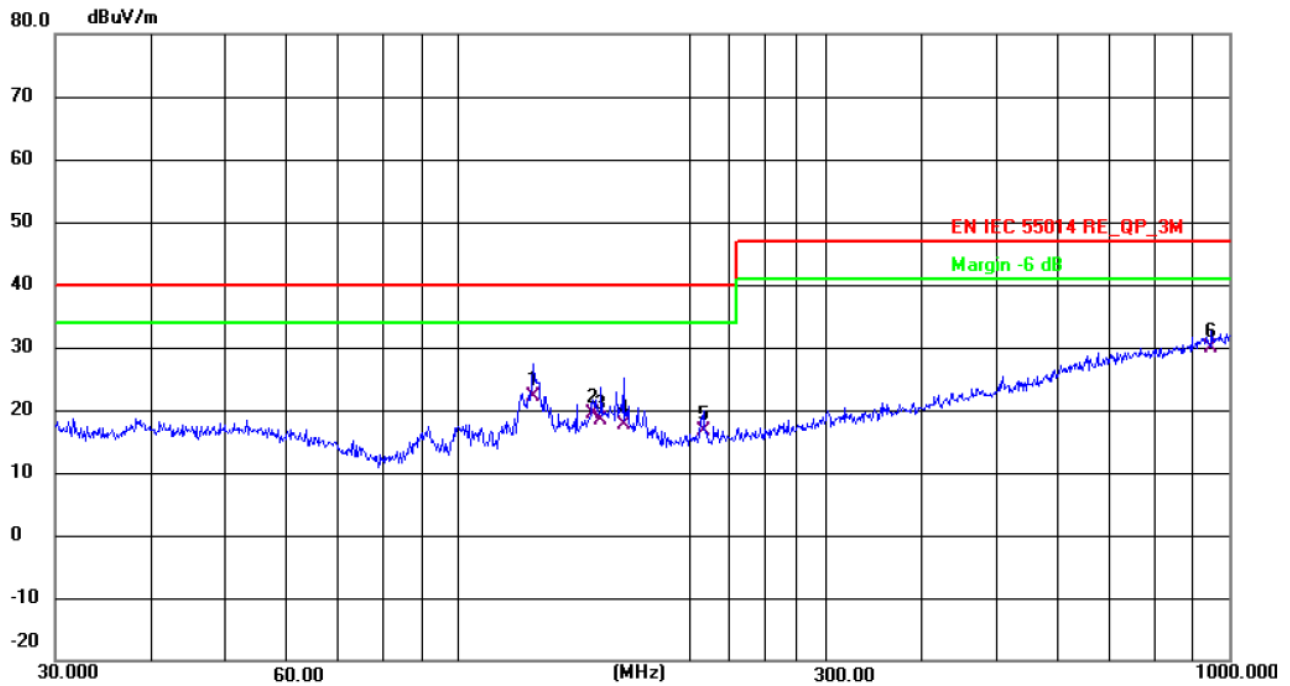
Remark: AC mode and DC mode were both tested, and most unfavorable test data of AC mode is recorded. And all the tests were carried out using AC/DC transformer for power supply.

Figure 5: Test Curve of Radiated Emission in the frequency range of 30-1000MHz, Horizontal (120V for Charging mode)



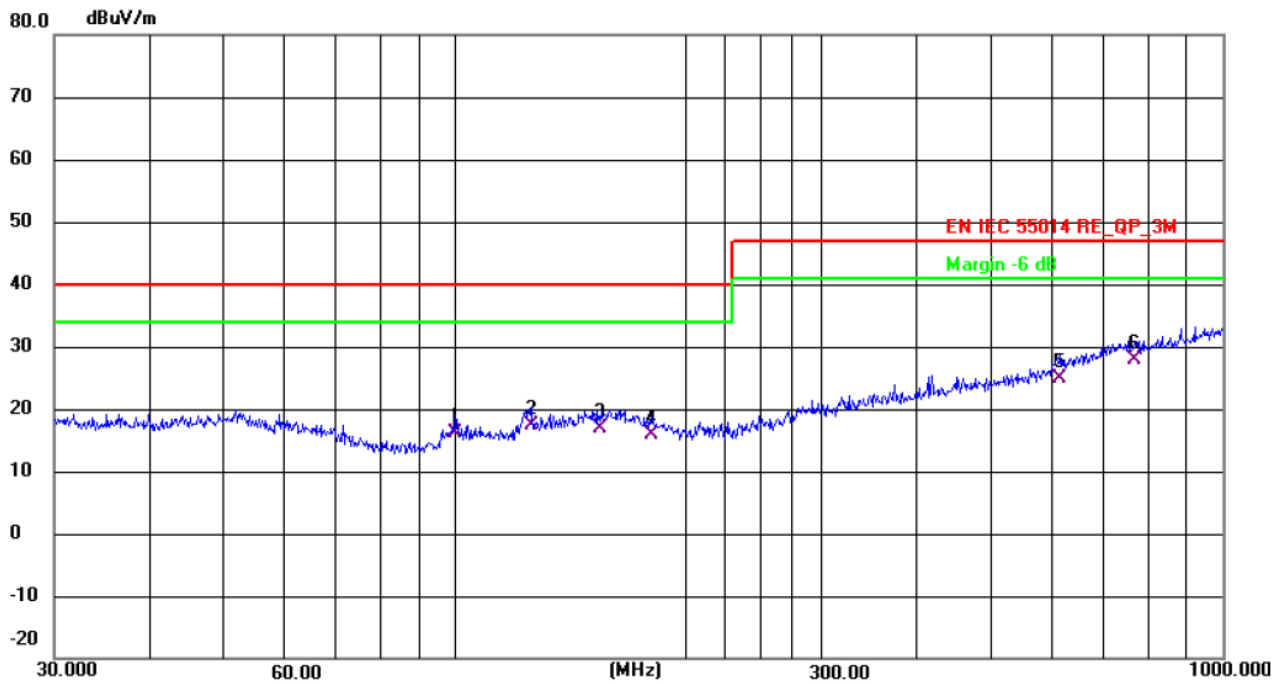
No.	Frequency (MHz)	Reading (dBuV)	Factor (dB/m)	Level (dBuV/m)	Limit (dBuV/m)	Margin (dB)	Detector	Height (cm)	Azimuth (deg.)	P/F	Remark
1	103.4421	3.57	11.23	14.80	40.00	-25.20	QP	400	72	P	
2	132.2206	3.21	13.39	16.60	40.00	-23.40	QP	100	178	P	
3	157.0074	3.18	14.32	17.50	40.00	-22.50	QP	100	106	P	
4	179.3863	2.42	13.48	15.90	40.00	-24.10	QP	200	232	P	
5	747.4825	3.95	24.45	28.40	47.00	-18.60	QP	200	169	P	
6 *	968.9338	3.77	26.33	30.10	47.00	-16.90	QP	100	360	P	

Figure 6: Test Curve of Radiated Emission in the frequency range of 30-1000MHz, Vertical (120V for Charging mode)



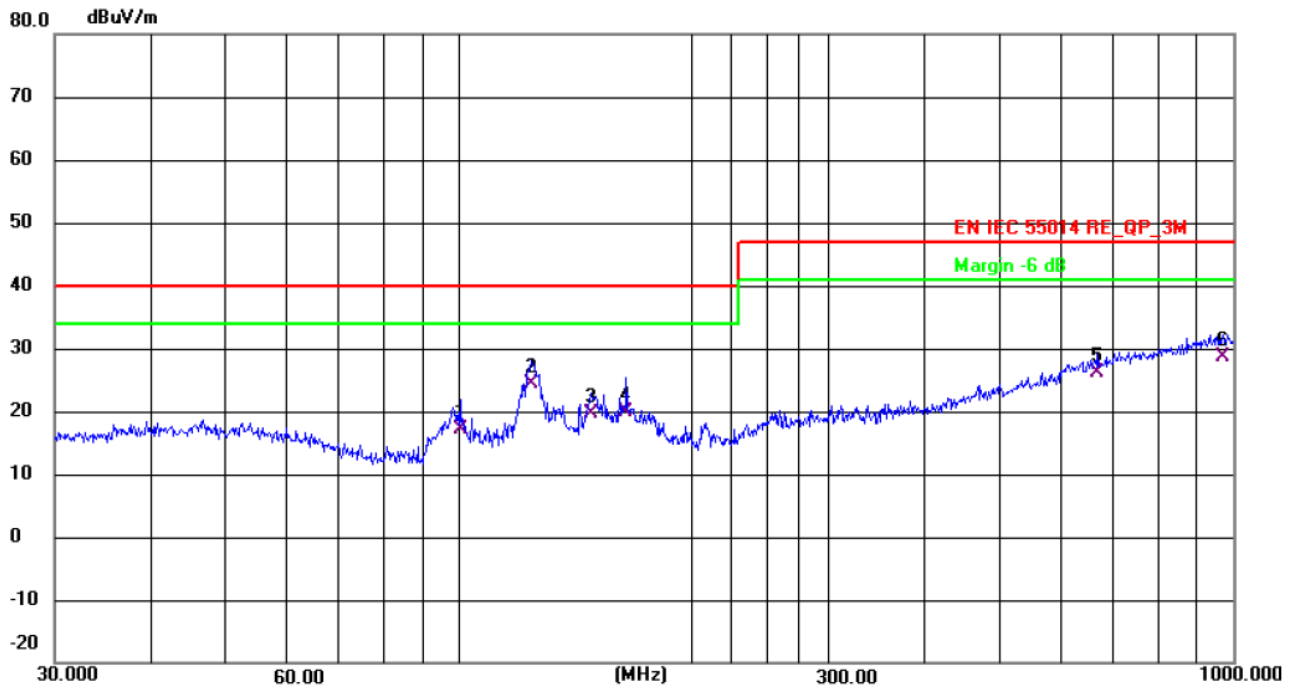
No.	Frequency (MHz)	Reading (dBuV)	Factor (dB/m)	Level (dBuV/m)	Limit (dBuV/m)	Margin (dB)	Detector	Height (cm)	Azimuth (deg.)	P/F	Remark
1	125.0066	10.35	11.85	22.20	40.00	-17.80	QP	100	132	P	
2	149.4857	5.75	13.55	19.30	40.00	-20.70	QP	200	360	P	
3	153.2004	4.92	13.58	18.50	40.00	-21.50	QP	100	313	P	
4	163.7550	4.35	13.25	17.60	40.00	-22.40	QP	100	325	P	
5	207.8501	6.74	9.96	16.70	40.00	-23.30	QP	300	165	P	
6 *	948.7610	4.35	25.45	29.80	47.00	-17.20	QP	100	203	P	

Figure 7: Test Curve of Radiated Emission in the frequency range of 30-1000MHz, Horizontal (230V for Charging mode)



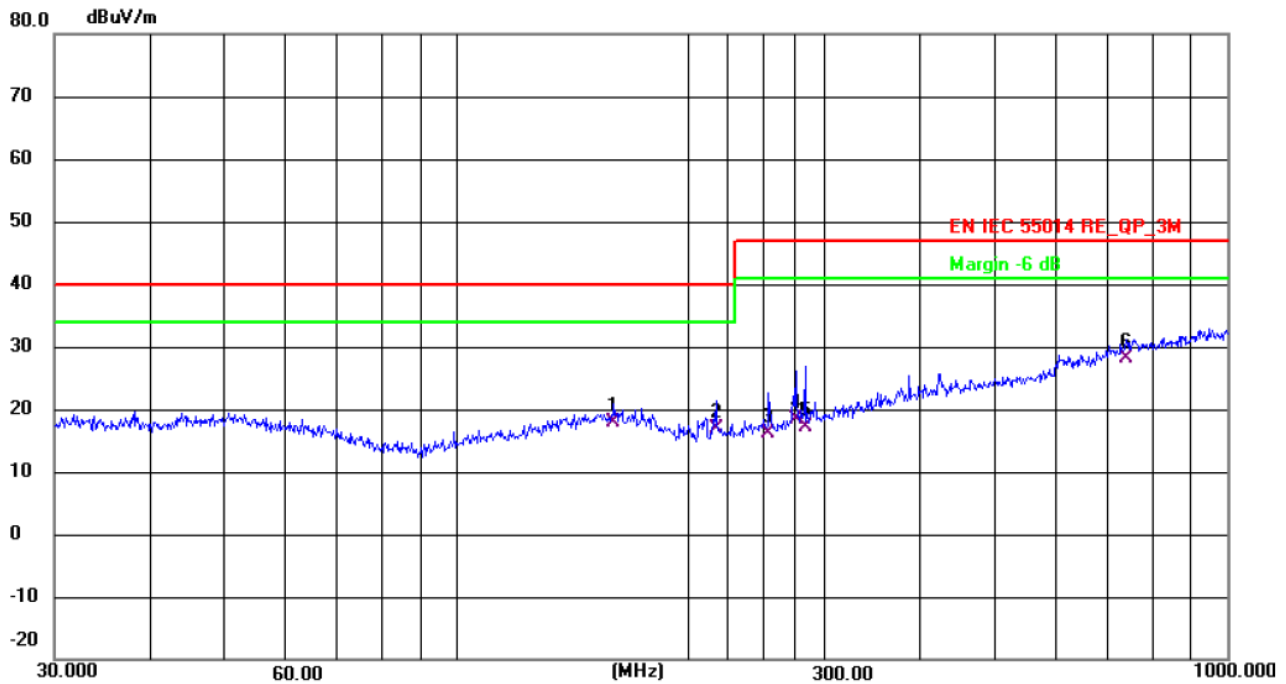
No.	Frequency (MHz)	Reading (dBuV)	Factor (dB/m)	Level (dBuV/m)	Limit (dBuV/m)	Margin (dB)	Detector	Height (cm)	Azimuth (deg.)	P/F	Remark
1	99.8777	5.32	10.83	16.15	40.00	-23.85	QP	300	239	P	
2	125.4457	4.67	12.73	17.40	40.00	-22.60	QP	100	344	P	
3	154.8204	2.51	14.29	16.80	40.00	-23.20	QP	100	167	P	
4	180.0165	2.57	13.43	16.00	40.00	-24.00	QP	200	273	P	
5	614.2142	2.96	21.94	24.90	47.00	-22.10	QP	100	315	P	
6 *	768.7481	3.39	24.51	27.90	47.00	-19.10	QP	100	78	P	

Figure 8: Test Curve of Radiated Emission in the frequency range of 30-1000MHz, Vertical (230V for Charging mode)



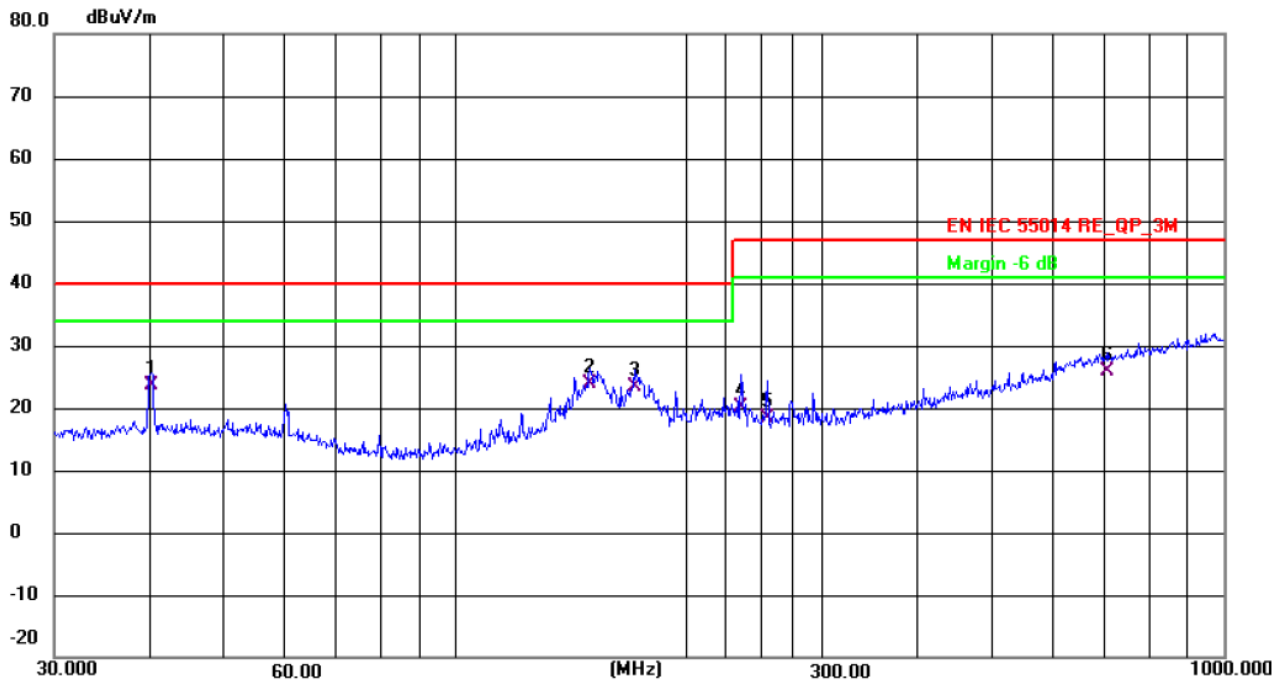
No.	Frequency (MHz)	Reading (dBuV)	Factor (dB/m)	Level (dBuV/m)	Limit (dBuV/m)	Margin (dB)	Detector	Height (cm)	Azimuth (deg.)	P/F	Remark
1	100.2285	7.79	9.41	17.20	40.00	-22.80	QP	100	155	P	
2 *	123.6984	12.63	11.77	24.40	40.00	-15.60	QP	200	303	P	
3	147.9214	6.17	13.43	19.60	40.00	-20.40	QP	100	315	P	
4	164.3301	6.71	13.19	19.90	40.00	-20.10	QP	100	328	P	
5	668.1422	4.11	21.99	26.10	47.00	-20.90	QP	300	151	P	
6	968.9337	3.15	25.55	28.70	47.00	-18.30	QP	100	33	P	

Figure 9: Test Curve of Radiated Emission in the frequency range of 30-1000MHz, Horizontal (Normal operation)



No.	Frequency (MHz)	Reading (dBuV)	Factor (dB/m)	Level (dBuV/m)	Limit (dBuV/m)	Margin (dB)	Detector	Height (cm)	Azimuth (deg.)	P/F	Remark
1	159.7844	3.44	14.36	17.80	40.00	-22.20	QP	200	53	P	
2	216.7828	5.36	11.44	16.80	40.00	-23.20	QP	100	218	P	
3	253.8367	3.26	12.94	16.20	47.00	-30.80	QP	100	206	P	
4	275.1570	4.46	13.84	18.30	47.00	-28.70	QP	300	201	P	
5	282.9852	2.96	14.14	17.10	47.00	-29.90	QP	100	36	P	
6 *	739.6604	3.86	24.34	28.20	47.00	-18.80	QP	100	135	P	

Figure 10: Test Curve of Radiated Emission in the frequency range of 30-1000MHz, Vertical (Normal operation)



No.	Frequency (MHz)	Reading (dBuV)	Factor (dB/m)	Level (dBuV/m)	Limit (dBuV/m)	Margin (dB)	Detector	Height (cm)	Azimuth (deg.)	P/F	Remark
1	40.1347	10.52	13.08	23.60	40.00	-16.40	QP	100	235	P	
2 *	149.4857	10.25	13.55	23.80	40.00	-16.20	QP	400	129	P	
3	170.7926	10.94	12.56	23.50	40.00	-16.50	QP	200	82	P	
4	234.9909	8.52	11.58	20.10	47.00	-26.90	QP	100	146	P	
5	254.7284	6.19	12.11	18.30	47.00	-28.70	QP	200	197	P	
6	706.6999	3.49	22.51	26.00	47.00	-21.00	QP	100	48	P	

5 Test Results I M M U N I T Y

According to the electrical characteristics above and EN IEC 55014-2:2021, the EUT belongs to category II (Charging mode) and III (normal operation) equipment.

“Category I: equipment containing no electronic control circuitry.”

“Category II: mains operated equipment containing electronic control circuitry with no clock frequency higher than 15 MHz.”

“Category III: battery operated equipment not included in Category I.”

“Category IV: mains operated equipment containing electronic control circuitry with a highest clock frequency greater than 15 MHz but lower than or equal to 200 MHz.”

“Category V: mains operated equipment containing electronic control circuitry with a highest clock frequency greater than 200 MHz.”

Performance criterion:

Performance criterion A: The apparatus shall continue to operate as intended during the test. No degradation of performance or loss of function is allowed below a performance level (or permissible loss of performance) specified by the manufacturer, when the apparatus is used as intended.

Performance criterion B: The apparatus shall continue to operate as intended after the test. No degradation of performance or loss of function is allowed below a performance level (or permissible loss of performance) specified by the manufacturer, when the apparatus is used as intended. During the test, degradation of performance is allowed.

Performance criterion C: Temporary loss of function is allowed, provided the function is self-recoverable or can be restored by the operation of the controls, or by any operation specified in the instructions for use.

Room temperature	:	24-26 °C
Relative Humidity	:	45-58%

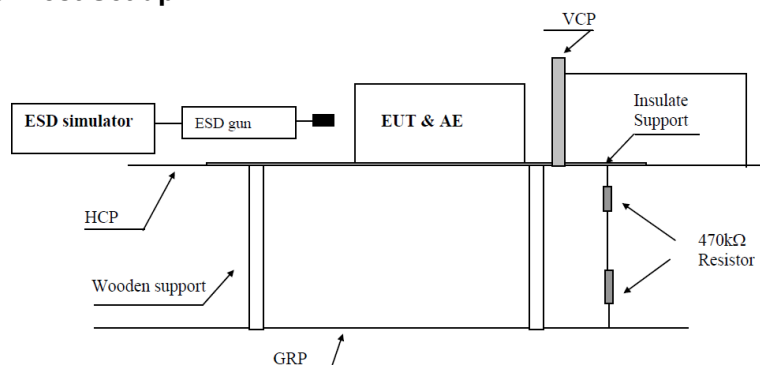
Conclusion: Pass

5.1 Enclosure

5.1.1 Electrostatic Discharge

Charge voltage	: $\pm 4.0\text{kV}$ (Conducted Discharge) $\pm 8.0\text{kV}$ (Air Discharge)
Polarity	: positive / negative
Number of discharges	: >10
Operational condition	: Charging mode and normal operation
Performance criteria	: B

Block Diagram of Test Set up



- ☒ For table top equipment, wooden support is 0.8m height.
- ☐ For floor standing equipment, wooden support is 0.1m height.

Test Procedure

The immunity against electrostatic discharge was tested in accordance with EN IEC 55014-2:2021. Test setup and ESD-Generator are according to EN 61000-4-2 which is specified by EN IEC 55014-2:2021.

The EUT is placed on 0,8m wood table above the ground plane. And the minimum distance between the EUT and all other conductive structures except the ground plane beneath the EUT is more than 0,5m. The reference ground plane is an aluminum sheet of 0,25mm minimum thickness. The reference ground plane is connected to the protective earth. The size of the ground plane is 2m × 2m.

A horizontal coupling plane (HCP), 1,6m × 0,8m, is placed on the table and isolated from the EUT and cables by an insulating support 0,5mm thick. Vertical coupling plane (VCP) of dimensions 0,5m × 0,5m is placed parallel to and positioned at a distance of 0,1m from the EUT.

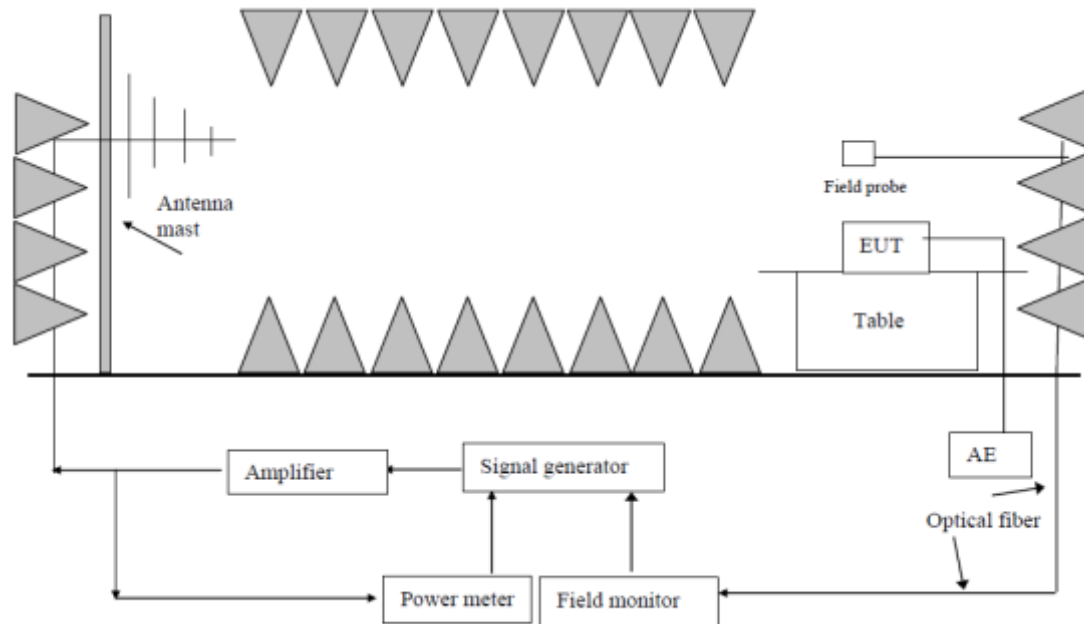
Table 3: ESD, Positive / Negative Polarity

Position	Kind of Discharge	Remarks	Result
Accessible nonmetal Enclosure	Air discharge $\pm 8\text{kV}$	No change of function	Pass
Metal Enclosure	Contact discharge $\pm 4\text{kV}$	No change of function	Pass
Coupling plane (Both HCP and VCP)	Contact discharge $\pm 4\text{kV}$	No change of function	Pass

5.1.2 Radio Frequency Electromagnetic Field

Test Level	: 3V/m
Frequency Range	: 80-1000MHz
Modulation	: 80%AM, 1kHz
Frequency Sweep Speed	: 0.005 octave/s (1.5×1E-3 decades/s)
Position	: Each side of the EUT
Operational condition	: Charging mode and normal operation
Performance Criteria	: A

Block Diagram of Test Set up



Test Procedure

The immunity against radio-frequency electromagnetic fields in the frequency range between 80MHz to 1000MHz was tested in accordance to IEC 61000-4-3 which is specified by Table 11 in EN IEC 55014-2:2021.

The test was performed inside a 3m modified semi-anechoic chamber. During the test the part of the ground plane between the field generating antenna and the equipment under test was covered by absorbing material. The distance between the tip of the antenna and the side of the system tested is 3m. The field uniformity of the 1.5mx1.5m plane where the surface of the EUT tested coincides with is regularly calibrated to ensure the 0-6dB field uniformity criterion as specified by IEC 61000-4-3 is met.

Table 4: Radiated Susceptibility, Field Strength 3V/m, Frequency 80MHz to 1000MHz

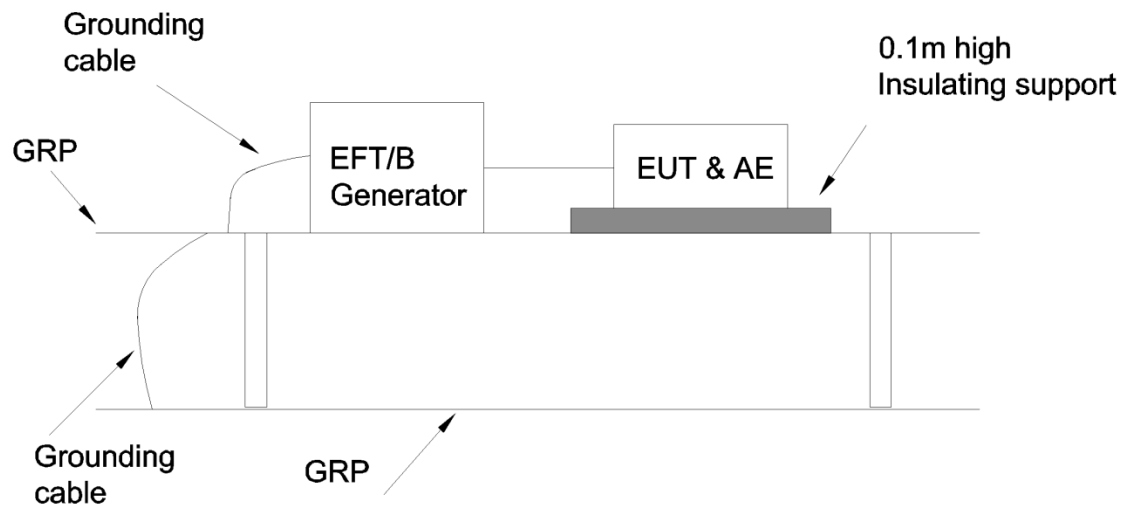
Position	Result	Remarks
Antenna in vertical orientation	Pass	No disturbance of function
Antenna in horizontal orientation	Pass	No disturbance of function

5.2 Input and Output AC Power Ports

5.2.1 Fast Transients on AC Power Lines

Test Voltage	:	$\pm 1\text{kV}$
Polarity	:	negative/positive
Repetition frequency	:	5kHz
Test duration	:	$\geq 120\text{sec}$
Tr/Tn	:	5ns/50ns
Operational condition	:	Charging mode
Performance criteria	:	B

Block Diagram of Test Set up



Test Procedure

The immunity against fast transients on AC power lines was tested in accordance to EN IEC 55014-2:2021. Test setup and the fast transient noise generator are according to EN 61000-4-4 which is specified by EN IEC 55014-2:2021.

The EUT is placed on 0,1m wood table above the ground plane. And the minimum distance between the EUT and all other conductive structures except the ground reference plane is more than 0,5m.

The length between the coupling device and the EUT is less than 1m. The cord length more than 1m, the excess length of the cable shall gather into a flat coil with a 0,4m diameter, and situated at a distance of 0,1m above the ground reference plane.

The reference ground plane is an aluminum sheet of 0,25mm minimum thickness. The reference ground plane is connected to the protective earth. The size of the ground plane is 2m × 2m.

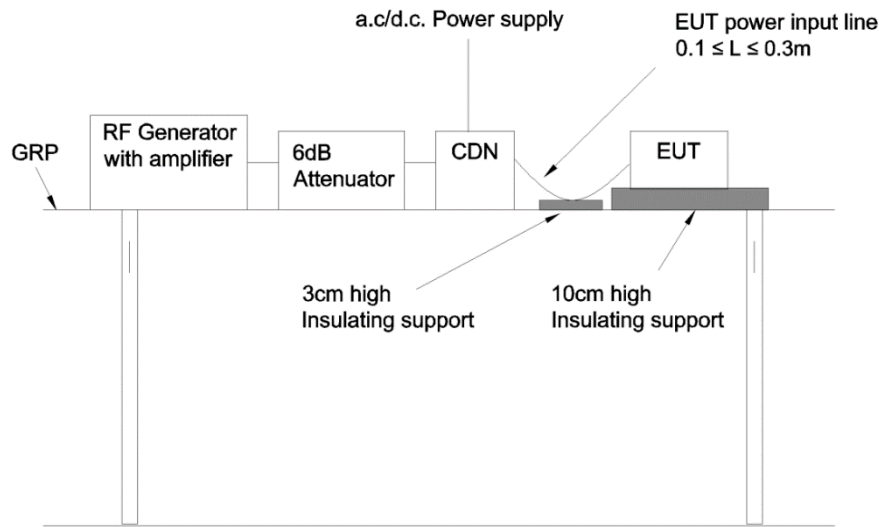
Table 5: Burst, AC Power lines, Positive and Negative Polarity

Line	Test Voltage / Result	Remark
AC Input (L+N)	$\pm 1\text{kV}$ Pass	No disturbance of function

5.2.2 Injected Current into AC Power Port

Voltage Level	: 3V(rms)(unmodulated)
Environmental phenomena	: r.f. current, common mode, 1kHz, 80%AM
Source impedance	: 150Ω
Frequency range	: 0.15-230 MHz
Sweeping rate	: $\leq 1,5 \times 10^{-3}$ decades/s
Operational condition	: Charging mode
Performance criteria	: A

Block Diagram of Test Set up



Test Procedure

The immunity against injected current into AC power port was tested according to EN IEC 55014-2:2021 in a shielded room. The Test setup and the test generator are according to EN 61000-4-6 which is specified by EN IEC 55014-2:2021.

The EUT is placed on 0,1m wood table above the ground plane. And the minimum distance between the EUT and all other conductive structures except the reference ground plane is more than 0,5m. The EUT comprised a single unit. The coupling and decoupling networks were inserted on the power supply connection. The coupling and decoupling networks was placed on the ground reference plane, making direct contact with it at about 0,1-0,3 meter from EUT. The cable between EUT and CDN is as short as possible and not bundled nor wrapped. The height of cable between the EUT and the coupling and decoupling networks above the ground reference plane was 50mm.

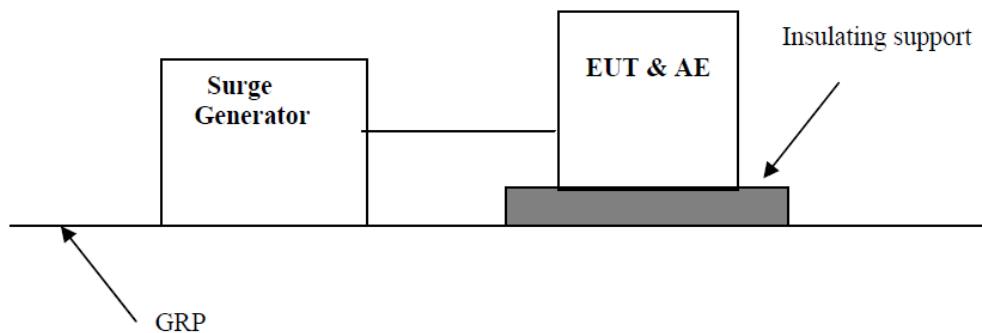
Table 6: Injected current, AC Power Port

Line	Coupling Method:	Remark	Result
AC Power Lines	CDN M-3	No disturbance of function	Pass

5.2.3 Surges to AC Power Port

Test Level	:	phase to neutral $\pm 1\text{kV}$ phase/neutral to PE $\pm 2\text{kV}$
Tr/Tn	:	1.2/50 μs (open-circuit voltage) 8/20 μs (short-circuit current)
Test numbers	:	5 positive and 5 negative pulses
Repetition rate	:	1 surge/min
Operational condition	:	Charging mode
Performance criteria	:	B

Block Diagram of Test Set up



Test Procedure

The immunity against surges to AC power port was tested in accordance to EN IEC 55014-2:2021. Test setup and the Combination Wave Generator (CWG) are according to EN 61000-4-5 which is specified by EN IEC 55014-2:2021. The EUT is placed on 0,1m wood table above the ground plane.

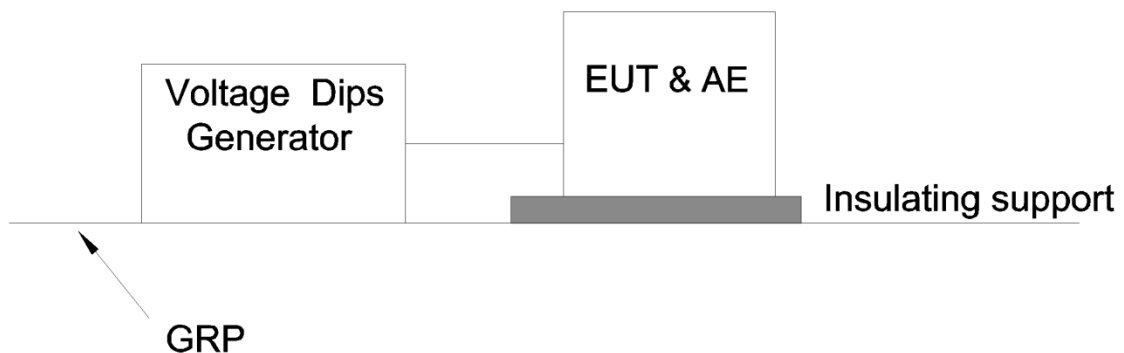
Table 7: Surges to AC Power lines, positive/negative

Line	Tested voltage/coupling phase	Test angle	Observation	Result
Phase to neutral	+1 kV, $+\pi/2$ (5 times) -1 kV, $-\pi/2$ (5 times)	90° 270°	No disturbance of function	Pass Pass

5.2.4 Voltage dips and interruptions to AC Power Port

Performance criteria	:	C
Test level (in % UT) and	:	0 0.5/0.5 periods(50/60Hz)
duration (in periods of the	:	40 10/12 periods(50/60Hz)
rated frequency)	:	70 25/30 periods(50/60Hz)
Operational condition	:	Charging mode

Block Diagram of Test Set up



Test Procedure

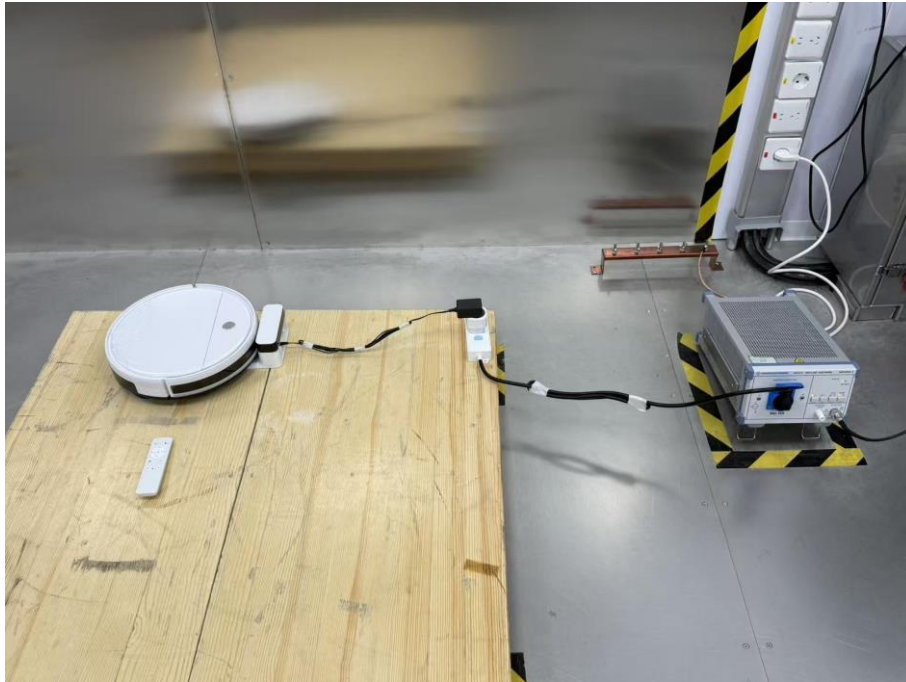
The immunity against voltage dips and interruptions to AC power port was tested in accordance to EN IEC 55014-2:2021. Test setup and the test generator are according to EN 61000-4-11 which is specified by EN IEC 55014-2:2021. The EUT was placed directly on the table of aluminum.

Table 8: Test condition and Test Result for Voltage dips and short interruptions

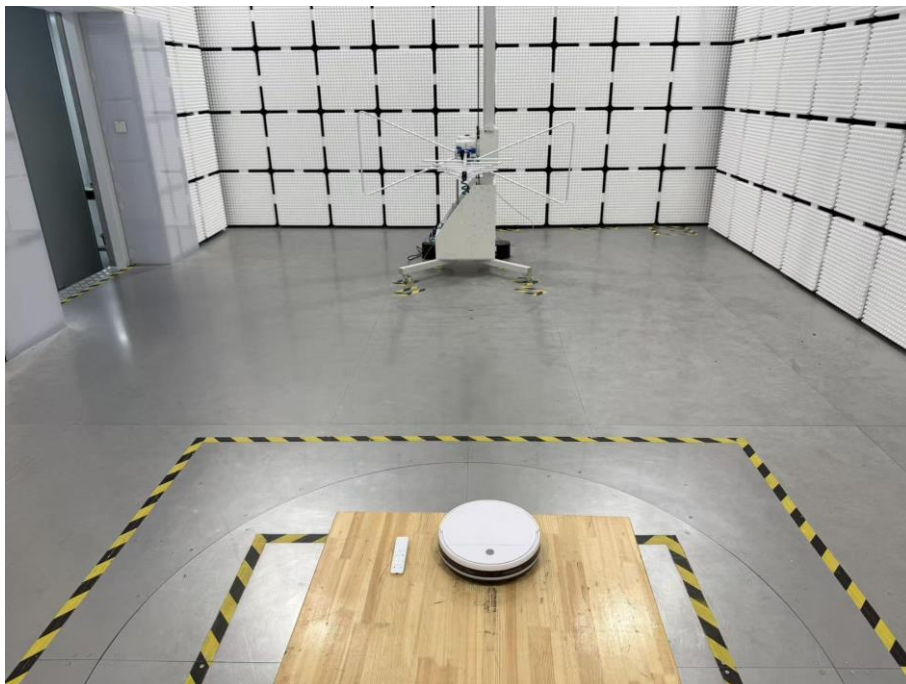
Test level (in % UT)	Duration	Performance criteria	Remarks	Result
0	0,5 (10ms)	C	No disturbance of function	Pass
40	10 (200ms)	C	No disturbance of function	Pass
70	25 (0.5s)	C	No disturbance of function	Pass

6 Photographs of the EUT

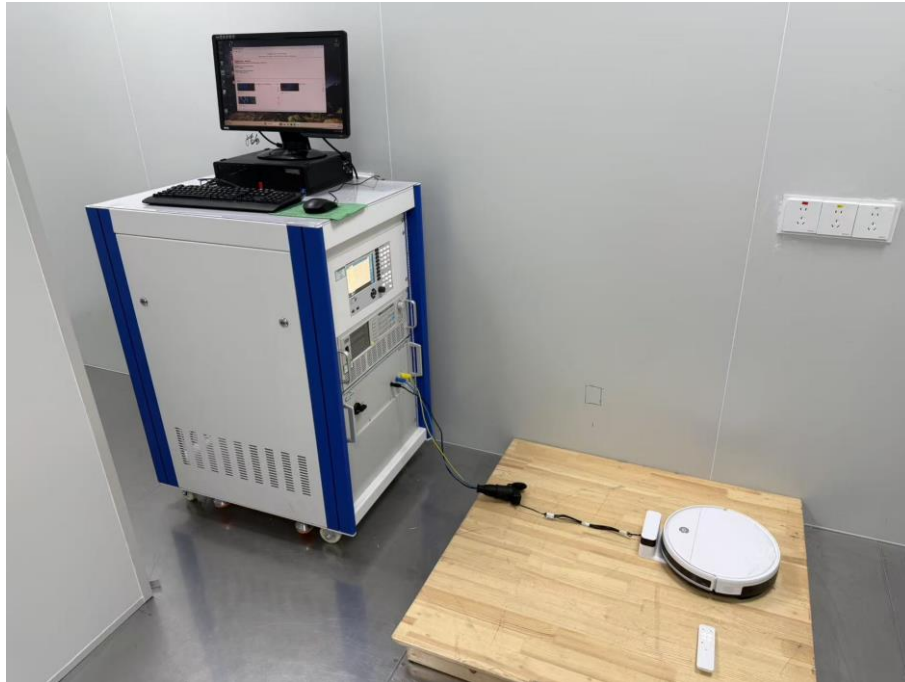
Photograph 1: Set-up for Disturbance Voltage



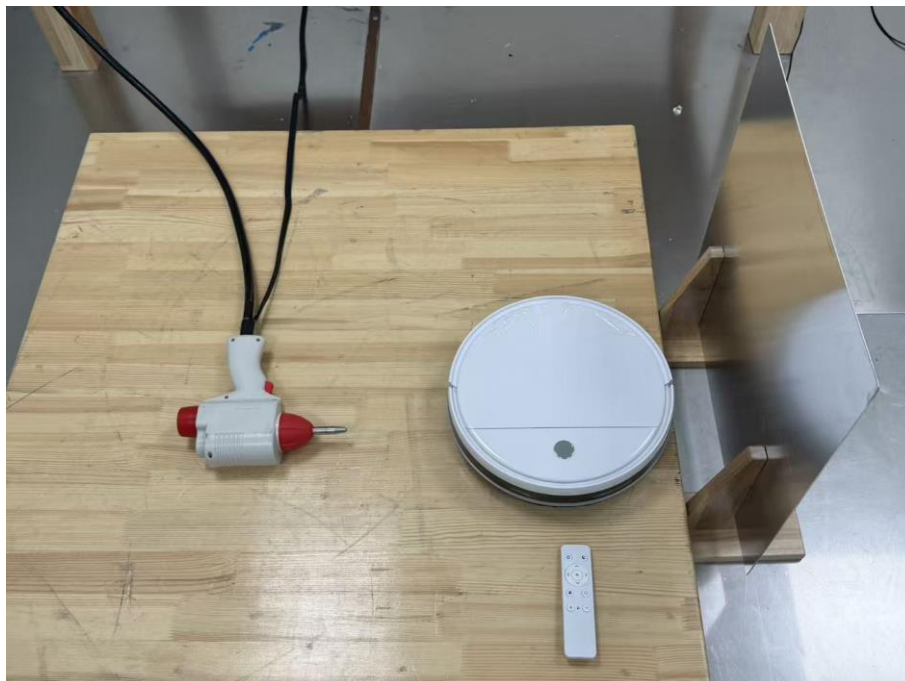
Photograph 2: Set-up for Radiated Emission



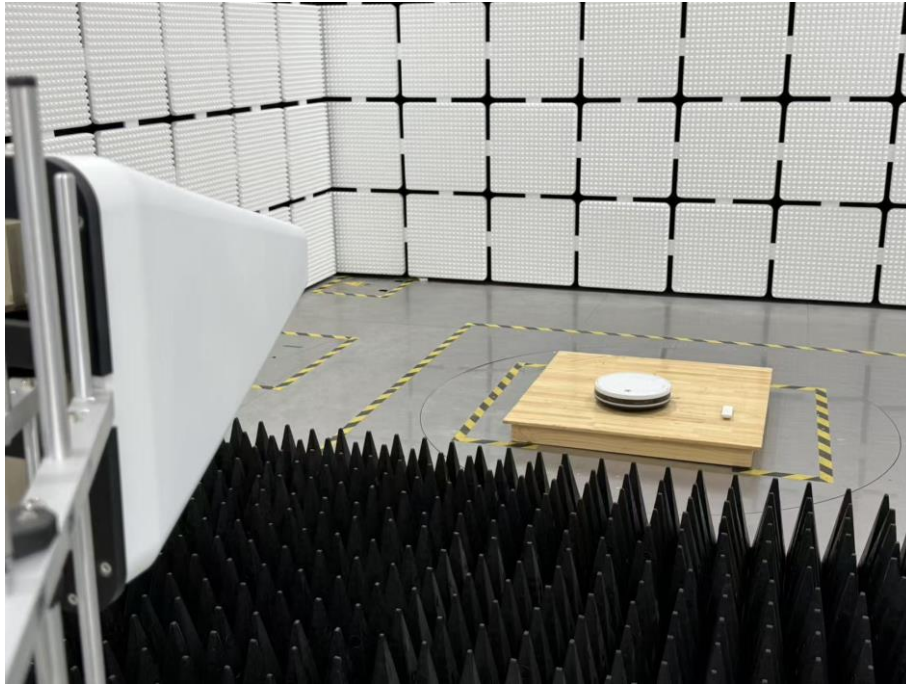
Photograph 3: Set-up for Flicker



Photograph 4: Set-up for ESD



Photograph 5: Set-up for Radiated Susceptibility



Photograph 6: Set-up for EFT, Surge and Voltage Dips



Photograph 7: Set-up for Injected Current



Photograph 8: Overall view of EUT



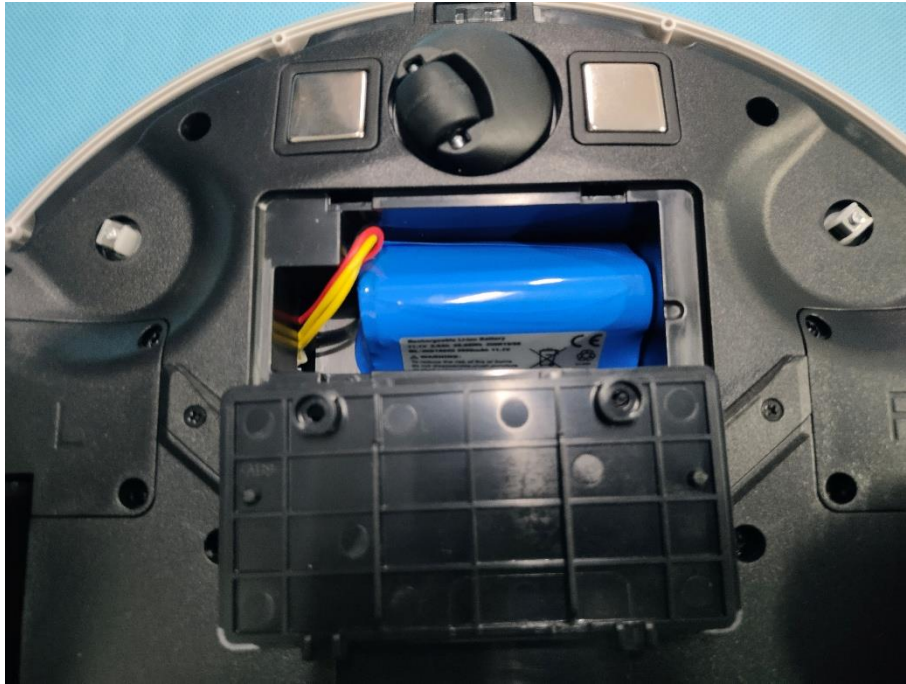
Photograph 9: Overall view of EUT



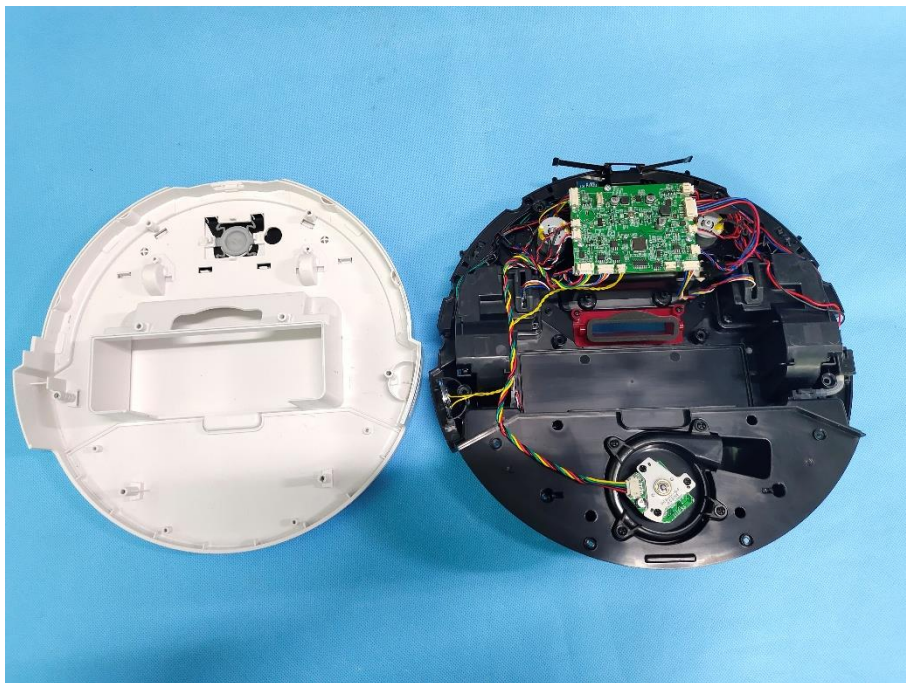
Photograph 10: Internal view of EUT



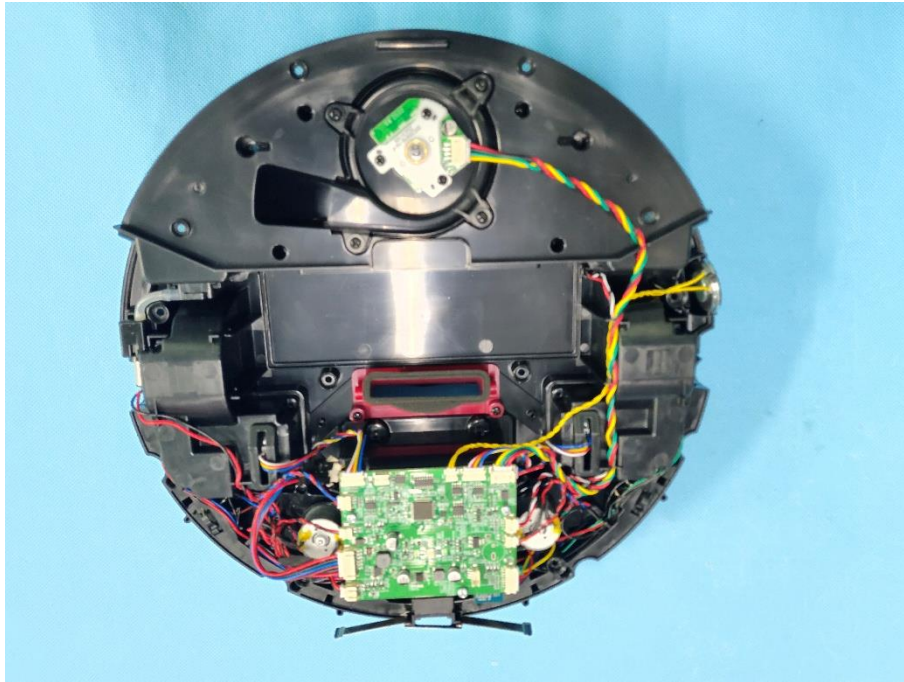
Photograph 11: Internal view of EUT



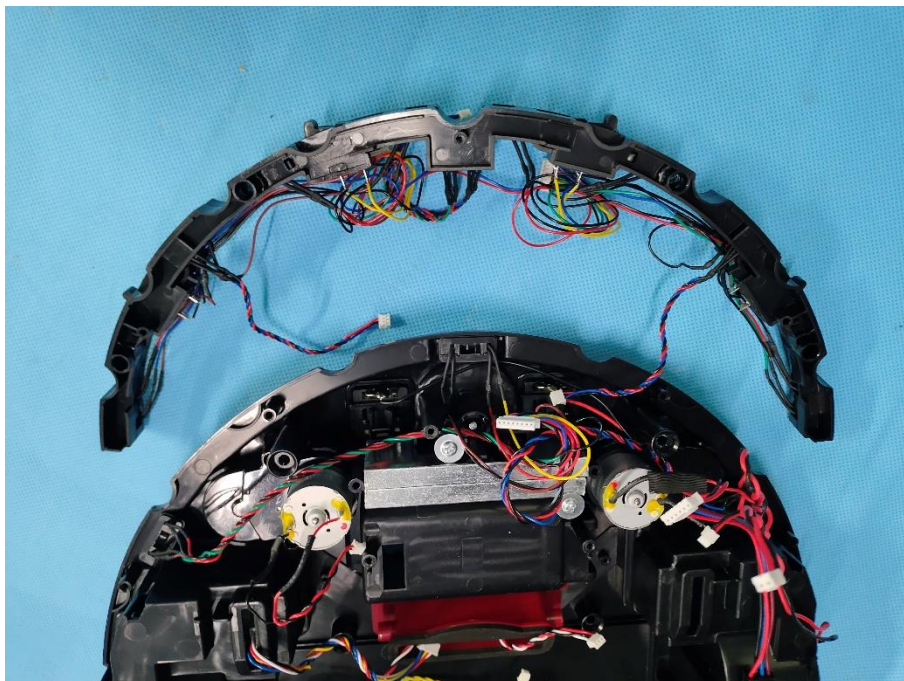
Photograph 12: Internal view of EUT



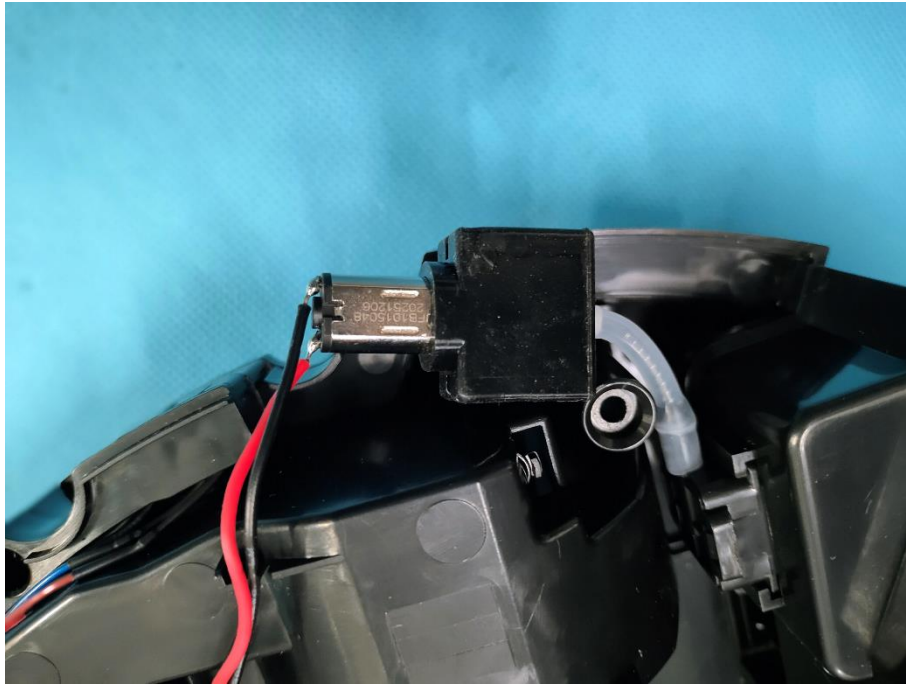
Photograph 13: Internal view of EUT



Photograph 14: Internal view of EUT



Photograph 15: Internal view of EUT



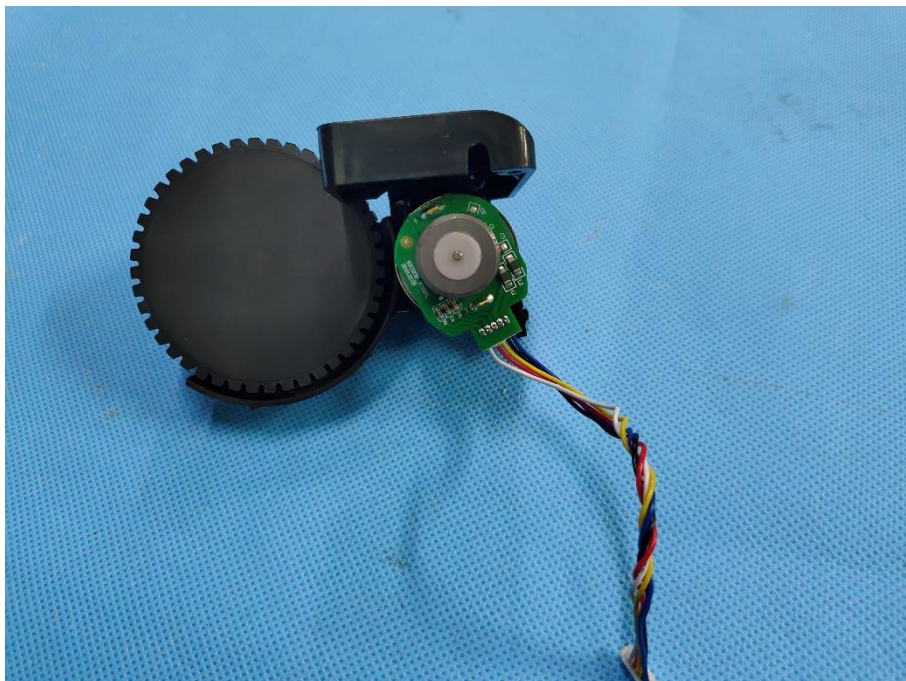
Photograph 16: Internal view of EUT



Photograph 17: Internal view of EUT



Photograph 18: Internal view of EUT



Photograph 19: Internal view of EUT



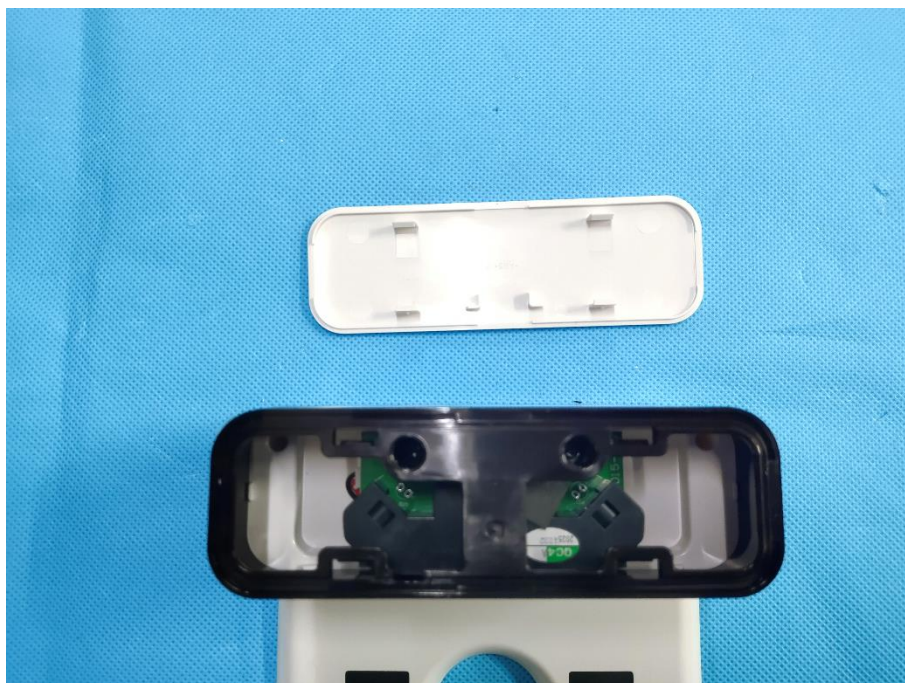
Photograph 20: Internal view of EUT



Photograph 21: Internal view of EUT



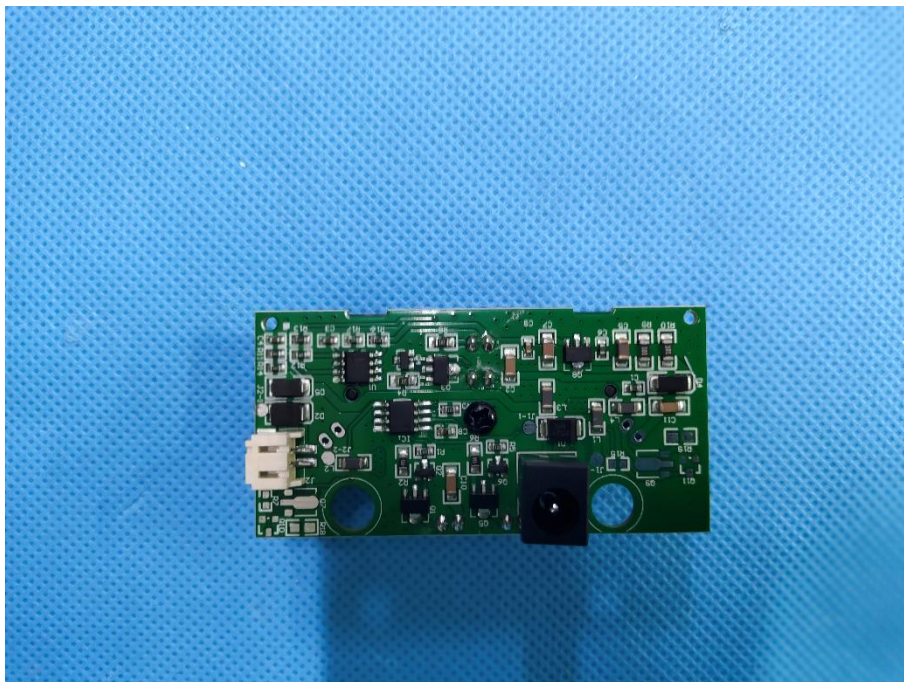
Photograph 22: Internal view of EUT



Photograph 23: Internal view of EUT



Photograph 24: Internal view of EUT



Photograph 25: Internal view of EUT



Photograph 26: Adapter view of EUT



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